
Fundamentals of Database

IT403

Smart Clinic Database System

Student Details:

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Instructions:

- You must submit two separate copies (**one Word file and one PDF file**) using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g. misspell words, remove spaces between words, hide characters, use different character sets, **convert text into image** or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You **MUST** show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

Project Instructions

- You can work on this project as a group (minimum 3 and maximum 4 students). Each group member must submit the project individually with all group member names mentioned on the cover page.
- This project **worth 20 marks** and will be distributed as in the following:
 - Database Design. (6 marks)
 - Database Implementation. (6 marks)
 - SQL Operations. (6 marks)
 - Project Reflection. (2 marks)
- Each leader in the group must submit one report about their chosen Project via the Blackboard **(Email submission will not be accepted and will be awarded ZERO marks)** containing the following:
 - A. Task 1 : database design
 - B. Task 2: database implementation
 - C. All requested queries/results
 - D. Screenshots from MySQL (or any other software you use) of **all the tables** after population and **query results**.
 - E. Reflection (300–500 words) describing the team's experience, challenges, design decisions, and AI usage (if applicable).
- Submit one **SQL file** containing the complete database implementation, including database creation, tables, data insertion, required queries, VIEW, and TRIGGER.
- Submit a **GitHub** (or GitLab) repository containing the complete project with regular commits and meaningful commit messages.
- Submit a **progress report** describing the completed work, key decisions, challenges, future plan, and contribution of each group member.
- Include **links** to the live report document, GitHub/GitLab repository, and Google Colab notebook (if applicable) in the **Project Artifacts** section of the final report.
- You are advised to make your work clear and well presented; marks may be reduced for poor presentation. This includes filling in your information on the cover page.
- You **MUST** show all your work, and text **must not** be converted into an image unless specified otherwise by the question.
- **Late submission will result in ZERO marks being awarded.**
- The work should be your own. Copying **from students or other resources, including AI software, will result in ZERO marks.**

Learning

Outcome(s):

CLO1, CLO2, CLO3,

CLO4, CLO5

Project Description

Design and Implementation of a Smart Clinic Database System

A private clinic currently manages its patient information manually, making it difficult to organize appointments, treatments, and payments. The clinic has decided to develop a database system to improve data organization and retrieval.

Your team is required to design, implement, and test a complete relational database using the concepts learned throughout this course. The project should demonstrate your understanding of database design, ER/EER modeling, relational mapping, SQL implementation, and advanced SQL queries. The project must be completed using MySQL.

Task 1 – Database Design (6 Marks)

- Design a complete database for the clinic that includes at least six entities such as Patients, Doctors, Appointments, Treatments, Medicines, Payments, or any other appropriate entities.
- A complete ER/EER diagram.
- Primary and foreign keys.
- Appropriate relationships with cardinality.
- At least one specialization/generalization (EER feature).
- Any assumptions made during the design process.

Task 2 – Database Implementation (6 Marks)

Complete the following tasks using MySQL. Include screenshots of both your SQL code and the corresponding output (results) for each task:

- Create the database and all required tables.
- Use appropriate data types and constraints (PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, etc.).
- Insert at least 5 records into each main table.

Task 3 – SQL Operations (6 Marks)

Complete the following tasks using MySQL. For each task, include screenshots of both your SQL code and the corresponding output (results), along with a brief description (one to two sentences) explaining the purpose of the query and the information it is intended to retrieve.

- Write SELECT statements.
- Write JOIN queries.
- Write nested queries.
- Use aggregate functions with GROUP BY.
- Write UPDATE and DELETE statements.
- Create one VIEW.
- Create one TRIGGER.

Task 4 – Reflection (2 Marks)

Write a **300–500-word reflection** on your experience while completing this project by answering the following questions. Your responses should be clear, concise, and supported with specific examples from your own work.

1. Describe one specific challenge or obstacle you encountered during this project and explain in detail how you identified and resolved it.
2. Explain why you chose your specific approach/method/model over at least one alternative you considered, and what tradeoffs informed that decision.
3. If you used any AI tools during this project (e.g., ChatGPT, Claude, Copilot), disclose which tool(s), for what specific purpose (e.g., debugging, grammar checking, brainstorming), and how you verified or modified the output.
4. What would you do differently if you were to redo this project with more time or resources?

Deliverables

1. **Project Report (PDF)** – System description, ER/EER diagram, relational schema, SQL code, screenshots, and reflection.
2. **SQL Script (.sql)** – Complete SQL implementation of the project.
3. **GitHub Repository** – Repository link with the complete project and commit history.
4. **Mid-Project Progress Report** – Summary of progress, challenges, decisions, and team contributions.
5. **Project Artifacts**
 - Live report document (with edit history): [link]
 - GitHub/GitLab repository (with commit history): [link]
 - Google Colab notebook (if applicable): [link]

Smart Clinic Database System - Progress Report

1. Work Completed During This Period

During this stage of the project, our team successfully completed the majority of the required work according to the project specifications.

A. Database Design (Task 1)

- Carefully analyzed the project requirements and identified all required entities for the Smart Clinic Database System.
- Designed a complete Enhanced Entity Relationship (EER) model that accurately represents the clinic's business processes.
- Implemented a specialization/generalization hierarchy where **Employee** serves as the superclass and **Doctor**, **Nurse**, and **Receptionist** are subclasses.
- Defined all entity attributes along with appropriate Primary Keys and Foreign Keys.
- Established all required relationships and cardinality constraints between entities.
- Converted the EER model into a normalized relational schema following Third Normal Form (3NF) to reduce redundancy and improve data consistency.

B. Database Implementation (Task 2)

The complete database has been implemented using MySQL.

The implementation includes:

- Creation of the **SmartClinic** database.
- Creation of all required tables using appropriate SQL statements.
- Definition of all Primary Keys.
- Definition of all Foreign Keys.
- Implementation of NOT NULL constraints.
- Implementation of UNIQUE constraints.
- Implementation of CHECK constraints where applicable.
- Proper use of AUTO_INCREMENT for identifier fields.
- Verification of referential integrity across all relationships.

The implemented database consists of the following tables:

- Employee
- Doctor
- Nurse
- Receptionist
- Department

- Patient
- Appointment
- Treatment
- Appointment_Treatment
- Medicine
- Prescription
- Prescription_Detail
- Payment

Sample data has been inserted into every main table, with at least five records where required by the project specification.

C. SQL Operations (Task 3)

The required SQL operations have been successfully implemented and tested.

Completed operations include:

- SELECT statements
- JOIN queries
- Nested (Subquery) statements
- Aggregate Functions with GROUP BY
- UPDATE statements
- DELETE statements
- VIEW creation
- TRIGGER creation

All SQL statements were executed successfully in MySQL and produced the expected results.

2. Key Design Decision

One of the most important design decisions made during this project was implementing a specialization/generalization hierarchy for employees.

Initially, the team considered storing all employee information inside a single Employee table. However, this approach would have required numerous nullable columns for role-specific information, resulting in redundant data and reduced normalization.

Instead, the team designed **Employee** as the superclass and **Doctor**, **Nurse**, and **Receptionist** as specialized subclasses.

This design provides several advantages:

- Eliminates redundant data.
- Improves database normalization.
- Preserves referential integrity.
- Makes future expansion easier.
- Accurately models the real-world organizational structure of the clinic.
- Fully satisfies the EER specialization/generalization requirement specified in the project instructions.

Although this design requires additional JOIN operations when retrieving employee information, the benefits in terms of scalability, maintainability, and data integrity significantly outweigh this minor complexity.

3. Challenge Encountered

One of the main challenges during the project was designing the many-to-many relationship between **Appointment** and **Treatment**.

Initially, it was unclear how multiple treatments could be assigned to a single appointment while also allowing the same treatment to be associated with multiple appointments.

After reviewing the database requirements and normalization principles, the team resolved this issue by introducing the **Appointment_Treatment** junction table.

This table uses a composite primary key consisting of:

- appointment_id
- treatment_id

This solution successfully eliminated data redundancy, preserved referential integrity, and allowed the database to support flexible treatment assignments while maintaining normalization.

Another challenge involved ensuring that all foreign key relationships were created in the correct order during database implementation. This was resolved by carefully organizing the table creation sequence and verifying all constraints before inserting sample data.

4. Remaining Work

At this stage, the database implementation is complete. The remaining tasks focus mainly on documentation and final project preparation.

The remaining work includes:

1. Capture screenshots of all populated database tables using MySQL Workbench.
2. Capture screenshots showing the execution and output of every required SQL query.
3. Insert the screenshots into the final project report with appropriate explanations.
4. Complete the Reflection section (300–500 words).
5. Upload the complete SQL script to the GitHub repository.
6. Verify that all project deliverables fully satisfy the assignment requirements.
7. Prepare the final PDF report for submission.

5. Group Members and Contributions

Student Name : Hanadi Abdullah Alshehri

- Database requirements analysis.
- Entity identification.
- ER/EER diagram design.
- Relationship and cardinality analysis.
- Relational schema preparation.

Student Name : Sakinah Ali Almuahini

- Database implementation in MySQL.
- Table creation.
- Primary and Foreign Key implementation.
- Constraint configuration.
- Data insertion and integrity testing.

Student Name : Sarah hadi alyami

- Development of SQL queries.
- Creation of SELECT, JOIN, Nested Queries, Aggregate Queries, UPDATE and DELETE statements.
- Development and testing of the VIEW.
- Development and testing of the TRIGGER.

Student Name : Kadi Alharbi

- Project documentation.
- Report formatting.
- Screenshot preparation.
- GitHub repository management.
- Final project review and quality assurance.

Overall Project Status

The Smart Clinic Database System has been successfully designed and implemented according to the project requirements. The database structure, SQL implementation, advanced SQL operations, VIEW, and TRIGGER have all been completed and tested successfully. The remaining work primarily involves documentation, screenshots, repository organization, and preparation of the final submission package.